

Agents Look Cooperative, Not Credible: Nonverbal Cues and the Credibility Gap in Human-Agent Teams

Mamehgo! Yousefi
School of Product Design
University of Canterbury
Christchurch, New Zealand
mahgol.yousefidashliboroun@pg.canterbury.ac.nz

Ahmad Shahi
Unitec
Auckland, New Zealand
sahmad@unitec.ac.nz

Ruoyu Wen
University of Canterbury
Christchurch, New Zealand
rwe77@uclive.ac.nz

Mos Sharifi
AgResearch
Lincoln, Canterbury, New Zealand
mos.sharifi@agresearch.co.nz

Alvaro J Jorge Romera
AgResearch
Christchurch, Canterbury, New Zealand
alvaro.romera@agresearch.co.nz

Simon Hoermann
School of Product Design
University of Canterbury
Christchurch, Canterbury, New Zealand
simon.hoermann@canterbury.ac.nz

Thammathip Piumsomboon
School of Product Design
University of Canterbury
Christchurch, Canterbury, New Zealand
tham.piumsomboon@canterbury.ac.nz

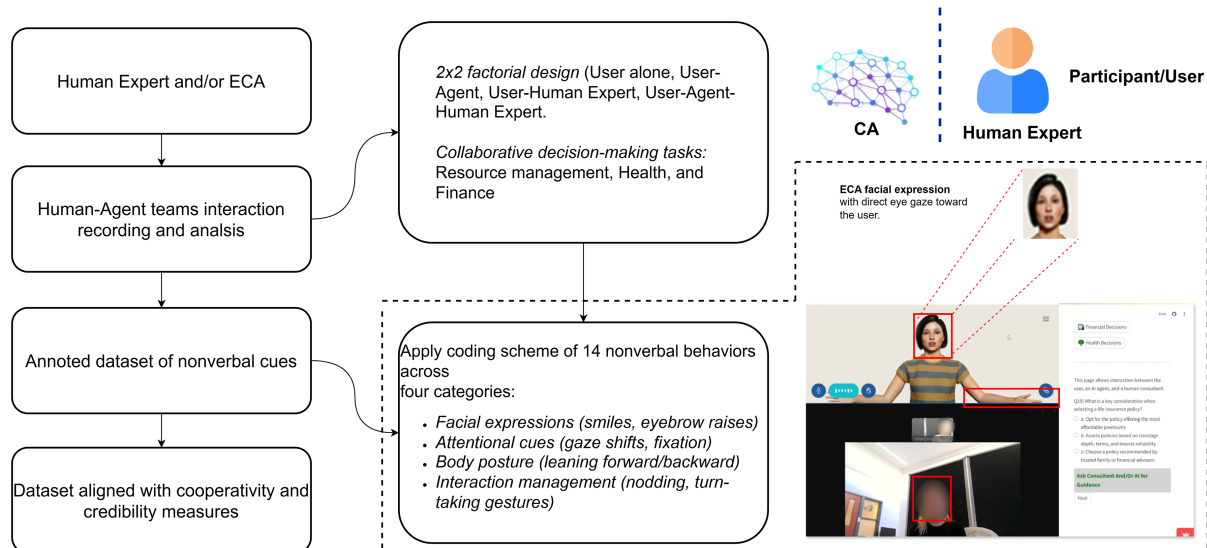


Figure 1: Study interface across four teammate configurations (UO, UA, UH, UAH) for decision-making tasks with on-demand advice.

Abstract

Embodied conversational agents (ECAs) increasingly mimic human nonverbal cues (NVCs), yet which discrete cues shape collaboration-critical perceptions in human-agent teams remains unclear. We report a within-subjects study ($N = 54$) spanning four teammate configurations: User Only (UO), User-Agent (UA), User-Human Expert (UH), and triadic User-Agent-Human Expert (UAH). From 1,356 minutes of recorded interaction, we coded 14 NVC categories



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(facial affect, gaze/attention, head/posture, and interaction management) with strong inter-rater reliability (Gwet's AC1 = 0.78–0.93) and modeled links to post-task ratings of perceived cooperativity and credibility using mixed-effects regression. Engagement cues, especially nodding, smiling, and turn-taking/backchannel reliably predicted higher perceived cooperativity for agents, including in UAH. Credibility was less responsive: identical signals (notably gaze and turn-taking) contributed more strongly when displayed by the human expert, indicating a persistent credibility gap. We outline implications for role-adaptive ECAs that pair cooperative signaling with epistemic transparency (e.g., provenance or uncertainty cues).

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Collaborative and social computing*; *Natural language interfaces*.

Keywords

embodied conversational agents, nonverbal cues, human-agent teaming, cooperativity, credibility, machine penalty

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1 Introduction

Recent advances in generative artificial intelligence (GenAI) have accelerated the adoption of conversational systems that support multimodal interaction (e.g., speech, text, and visual signals) in everyday collaborative tasks [6, 7, 12]. Embodied conversational agents (ECAs) powered by large language models (LLMs) can now produce rich nonverbal cues (NVCs) such as gaze shifts, nods, smiles, and turn-taking signals alongside fluent dialogue. These cues are central to socially intelligible interfaces because they regulate attention, convey engagement, and shape interpersonal impressions [2, 3, 5, 21]. Yet we still lack a systematic, cue-level understanding of which discrete NVCs shape collaboration, critical perceptions, especially cooperativity (willingness to work together) [9, 15] and credibility (perceived expertise and trustworthiness) [8] and whether the same cues are interpreted differently when enacted by an agent versus a human teammate.

Key contribution. We introduce a cue-level mapping that distinguishes cooperativity signaling from credibility attribution in both dyadic and triadic human-agent teams (HATs). This distinction matters because it reframes a common design assumption: making an agent feel cooperative is largely a social-signal problem, whereas making an agent seem credible is an epistemic-legibility problem that often cannot be solved by mimicry alone.

Background and gap. Prior work in ECA and HATs show that social signals can influence rapport, engagement, and trust-related judgments [4, 8, 13, 17, 20]. Yet two gaps remain. First, many studies treat embodiment or expressivity as broad constructs rather than isolating the contribution of fine-grained, discrete cues (e.g., nods

vs. gaze vs. smiles) during collaborative problem solving [1]. Second, evidence is dominated by dyadic settings; we know less about mixed, triadic collaboration where a user coordinates with both a human expert and an agent [10], and where attribution biases may shape whether NVCs translate into credibility (e.g., a persistent machine penalty).

In this paper, we address these gaps via a within-subjects study in which $N = 54$ participants completed structured decision-making tasks under four teammate configurations: User Only (UO), User-Agent (UA), User-Human Expert (UH), and User-Agent-Human Expert (UAH). Across all sessions, we captured 1,356 minutes of interaction video and coded 14 categories of teammate NVCs spanning facial expression, attention (gaze), posture/head movement, and interaction management (e.g., backchannels, turn-taking). We link these behaviors to participants' post-task ratings of perceived cooperativity and credibility using validated scales and mixed-effects models, enabling a cue-level comparison of how the same signals are weighted across agent vs. human teammates and dyadic vs. triadic team contexts.

Accordingly, the following research questions are elaborated as follows:

- RQ1:** How do specific agent nonverbal cues influence user perceptions of cooperativity in human-agent interaction (UA)?
- RQ2:** Which nonverbal cues most strongly predict perceived agent credibility in human-agent interaction (UA)?
- RQ3:** How do perceptions of cooperativity and credibility differ across teammate configurations (UO, UA, UH, UAH), and do identical cues carry different weight when enacted by agents versus humans?

We find a consistent split between cooperativity signaling and credibility attribution in human-agent collaboration. Engagement-signalling NVCs, especially nodding, smiling, turn-taking, and backchannels, robustly predict higher perceived cooperativity among agents in dyadic and triadic contexts. In contrast, credibility remains less responsive to agent NVCs, with identical signals (notably gaze and turn-taking) carrying significantly stronger weight when enacted by human experts, revealing a persistent "machine penalty." We contribute a fine-grained mapping from 14 discrete NVCs categories across within-subjects UA/UH/UAH conditions. We empirically demonstrate that while cooperativity signaling transfers effectively to embodied conversational agents, credibility attribution is systematically discounted for agents relative to humans. These findings yield actionable design guidance: role-adaptive agents should leverage coordinated engagement cues to signal cooperativity while pairing credibility-relevant moments with epistemic transparency mechanisms such as provenance information or calibrated uncertainty indicators to address the persistent credibility asymmetry.

2 Method

We report a within-subjects study and secondary analysis of multimodal recordings to test how discrete NVCs from an embodied conversational agent (ECA) and a human expert relate to perceived cooperativity and credibility across teammate configurations.

2.1 Design, tasks, and teammates

Study design followed a within-subjects design with team configuration as the primary independent variable, manipulating the presence or absence of an AI agent teammate and a human expert teammate, resulting in four conditions: User Only (UO), User-Agent (UA), User-Human Expert (UH), and User-Agent-Human Expert (UAH). Each participant completed decision-making tasks under all four configurations. In each session, participants made an initial decision and could revise it after soliciting input from available teammate(s). Configuration order was counterbalanced across task domains (Latin square) to reduce order/learning effects.

Participants. Fifty-four participants were recruited from a university community ($M_{age} = 26.25$, $SD_{age} = 6.00$). Participants were proficient English speakers and received compensation. The study protocol received ethics approval from the host institution's ethics committee.

Tasks & interaction mode. Tasks were implemented in a custom web app across three domains (resource management, finance, health) and used a Tree-of-Thought structure [23]. To preserve agency and reduce exposure differences between sources, the AI agent and/or human expert responded only when explicitly prompted (user-directed dialogue) [11].

Teammates and presentation. In UA/UAH, the ECA was a commercial digital human (Soul Machines HumanOS) [19] with responses generated by GPT-4o [16]; its NVCs (e.g., gaze, nods, smiles, gestures) were *not scripted/manipulated* and thus are analysed associatively. In UH/UAH, a researcher served as the human expert via video call in an adjacent window. Both teammates were presented alongside the task interface, providing comparable visual channels across conditions.

2.2 Measures and NVC coding

After each teammate-supported session, participants rated: Credibility (9 items; expertise, trustworthiness, goodwill; 7-point Likert; $\alpha = .89$) [14, 22] and Cooperativity (4 items; willingness, responsiveness, fairness, constructive contribution; 7-point Likert; $\alpha = .85$) [18]. We analysed 1,356 minutes of screen-recorded interaction and coded 14 discrete NVCs spanning facial affect, attentional cues, head/posture, and interaction management. Critically, cues were annotated with onset/offset and duration within problem-solving episodes, enabling features such as frequency and duration (e.g., gaze-to-participant duration; nod counts). Definitions followed a coding manual (e.g., turn-taking cue marked end-of-turn signals such as pause + gaze shift; backchannels marked brief listener signals during the partner's turn). We double-coded 20% of sessions; agreement was strong (Gwet's AC1=0.78–0.93) and disagreements were resolved by consensus.

2.3 Analysis and interpretation

We fit mixed-effects regression models predicting cooperativity and credibility from cue features with participant random intercepts. Fixed effects included teammate configuration and cue $\times$ configuration interactions to test whether identical cues were weighted differently across UA, UH, and UAH. We fit random-intercept models (no

random slopes) due to sample size; p-values are uncorrected and interpreted descriptively given multiple tests. Because teammate type also differs in framing and content (human expert vs. LLM-based agent), we interpret observed effects as associations and avoid causal claims about NVCs alone. To reduce systematic exposure differences, both sources responded only to explicit user prompts; however, we did not equate advice quality or authority framing. As a result, credibility effects may reflect both NVC weighting and differences in source framing or content.

3 Results

We examine how teammates' discrete NVCs relate to users' perceived cooperativity and credibility. Because each participant completed multiple configurations, we fit mixed-effects regression models with participant-level random intercepts. Predictors were standardised cue features derived from the coded cue stream within problem-solving episodes. Cue features were aggregated to the session level (participant $\times$ onfiguration) by summarising episode-level annotations within each session (e.g., mean duration and/or rate across that session's episodes), aligning predictors with the post-session cooperativity/credibility ratings. Predictors further included teammate configuration (UA, UH, UAH) and cue $\times$ configuration interactions to test whether the same cue is weighted differently for agents vs. humans and in dyads vs. triads. We report standardised coefficients (β) with conventional significance thresholds, and interpret cue-level effects as associational (not causal) and primarily descriptive, emphasising consistent patterns across configurations.

Scope. UO contains no teammate behavior and is therefore excluded from cue $\rightarrow$ perception models; all cue-based analyses use UA, UH, and UAH.

3.1 Cue matrix overview

Figure 2 summarises cue-level associations for both outcomes. Cooperativity shows broader and more consistently positive cue associations than credibility. Credibility effects concentrate in fewer cues and are systematically stronger for humans than agents for the same signals (notably gaze and turn-taking), consistent with a credibility-specific "machine penalty" (i.e., attribution differences rather than absence of agent signaling).

RQ1: Which agent cues predict perceived cooperativity?—Consistent with engagement-signaling accounts of cooperative intent, the strongest and most reliable predictors of cooperativity are expressive and interaction-management cues. In UA, agent *smiling* ($\beta = 0.31$, $p < .01$), *nodding* ($\beta = 0.28$, $p < .01$), and *turn-taking cues* ($\beta = 0.24$, $p < .05$) were positively associated with cooperativity (Figure 2). *Backchannels* also showed a positive association ($\beta = 0.20$, $p < .05$), reinforcing that responsiveness drives cooperative impressions. In the triadic UAH setting, *attention allocation* becomes especially salient: *agent gaze at the participant* was strongly associated with cooperativity ($\beta = 0.30$, $p < .01$).

RQ2: Which cues predict perceived credibility, and do humans vs. agents differ?—Credibility shows fewer and weaker cue-level associations than cooperativity. We interpret "not credible" as a relative credibility gap: agents benefit from some cues, but the same signals are weighted more strongly for humans. For the

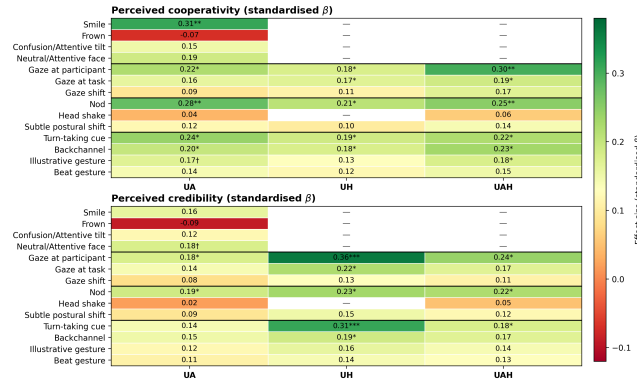


Figure 2: Cue matrix (standardised β) linking 14 NVCs to perceived cooperativity and credibility across UA/UH/UAH; green/red indicate positive/negative effects, and “—” marks not applicable. Significance: $^{\dagger}p < .10$, $^*p < .05$, $^{}p < .01$, $^{***}p < .001$.**

Table 1: Human vs. agent perception gap (UA vs. UH).

Measure	UA $M(SD)$	UH $M(SD)$	p	d
Cooperativity	4.92 (1.14)	5.78 (0.82)	$< .001$	0.87
Credibility	4.83 (1.25)	5.41 (0.97)	.006	0.52

Note. d is an approximate standardised mean difference using pooled SD.

agent in UA, nodding is the most consistent positive predictor of credibility ($\beta = 0.19$, $p < .05$), and neutral/attentive facial displays show only a marginal trend ($\beta = 0.18$, $p < .10$). Crucially, we observe a systematic HA asymmetry for credibility-relevant signals. In UH, gaze at the participant predicts credibility very strongly ($\beta = 0.36$, $p < .001$), exceeding the corresponding association for the agent in UA ($\beta = 0.18$, $p < .05$). Similarly, turn-taking cues are a strong credibility signal for humans ($\beta = 0.31$, $p < .001$) but comparatively weak for agents ($\beta = 0.14$, n.s.). We interpret these patterns as attribution differences consistent with a credibility-specific “machine penalty,” while noting that teammate type also differs in source framing and response generation (human expert vs. LLM-based agent), so effects should not be read as causal effects of NVCs alone.

RQ3: Do overall perceptions differ by teammate configuration?—Human experts received higher ratings than agents on both outcomes. Cooperativity was higher for humans in UH ($M = 5.78$, $SD = 0.82$) than for agents in UA ($M = 4.92$, $SD = 1.14$; $p < .001$). Credibility showed the same direction: UH ($M = 5.41$, $SD = 0.97$) exceeded UA ($M = 4.83$, $SD = 1.25$; $p = .006$). Table 1 summarises the gap with an approximate standardised mean difference.

4 Discussion

Our findings provide a cue-level account of how NVCs shape two collaboration-critical perceptions in human-agent teams: cooperativity and credibility. Across configurations, cooperativity was broadly responsive to engagement-signaling cues, whereas credibility was comparatively resistant for agents: the same signals

especially gaze and turn-taking were weighted more strongly when enacted by the human expert (Figure 2). This separation suggests a practical design split: cooperativity can often be supported via social coordination signals, while credibility requires epistemic legibility beyond mimicry. Because cues were naturally occurring and teammate type differs in both source framing and response generation (human expert vs. LLM-based agent), effects are interpreted as associational rather than causal [1, 4, 8]. Furthermore, the human expert was presented via live video with natural voice timing, while the ECA used a 3D avatar with synthesized speech and potential response latency. These differences in embodiment realism, audio quality, and interaction timing may independently influence credibility perceptions beyond NVC content alone. Accordingly, observed credibility asymmetries should be interpreted as reflecting a combination of NVC weighting, source framing, and modality-level differences, rather than being attributable to non-verbal cues alone. Nonetheless, the cross-condition consistency in Figure 2 provides actionable design guidance for embodied agents in human-agent teams.

Cooperativity is driven by engagement cues (RQ1). In UA, the strongest cooperativity associations were agent smiling ($\beta = 0.31$), nodding ($\beta = 0.28$), turn-taking cues ($\beta = 0.24$), and backchannels ($\beta = 0.20$), indicating that responsiveness and clear interaction management reliably signal willingness to collaborate (Figure 2). In triadic UAH, agent gaze-at-participant remained a salient cooperativity signal ($\beta = 0.30$), suggesting that attention allocation becomes especially important when users coordinate with multiple teammates.

Credibility shows an attribution asymmetry (RQ2). For the agent in UA, credibility benefited modestly from nodding ($\beta = 0.19$) and (marginally) neutral/attentive facial displays ($\beta = 0.18$), but overall credibility effects were fewer and weaker than cooperativity effects. By contrast, for the human expert in UH, gaze-at-participant was strongly associated with credibility ($\beta = 0.36$), and turn-taking cues were also a strong credibility signal ($\beta = 0.31$), exceeding the corresponding association for the agent (turn-taking: $\beta = 0.14$, n.s.). Together, these patterns indicate a credibility-specific attribution gap: cues that communicate competence and trustworthiness for humans are discounted when expressed by an agent.

Humans are rated higher overall (RQ3). Consistent with the above asymmetry, overall ratings favored the human expert over the agent for both cooperativity and credibility (Table 1), reinforcing that cooperative signals can transfer to agents, but credibility attribution remains harder to shift via NVCs alone.

Design implications.

(Cooperativity) Engineer responsiveness, not intensity.—To enhance perceived cooperativity, prioritise timing of NVCs display rather than frequency. Design agents to nod, backchannel, and smile in sync with user turns, use context-appropriate expressions, and signal clear turn-taking. This creates responsive, natural interactions rather than scripted ones.

(Credibility) Pair Nonverbal Cues with Epistemic Transparency.—Since credibility cues like gaze and turn-taking are weighted more strongly for humans than agents, supplement nonverbal signals

with explainability mechanisms at advice moments. Provide light-weight transparency features such as provenance information, uncertainty indicators, or brief rationale so users can verify the evidence behind the agent's displayed confidence.

(Multi-Party) *Make role and attention management explicit in triads*— Since agent gaze-at-participant strongly signals cooperativity in triad teams, implement role-aware gaze and floor control, displaying supportive listening (stable gaze toward speakers) during expert contributions, then shift to clear turn-taking and direct gaze when users address the agent. This helps users navigate complex interactions and understand the agent's participation mode.

5 Conclusion

We find that engagement-oriented NVCs robustly signal cooperativity for embodied agents, but credibility reveals a fundamental asymmetry: humans weight identical cues, particularly gaze and turn-taking more heavily when displayed by human versus agent teammates. This asymmetry suggests that cooperativity and credibility demand distinct design strategies. Cooperativity is a coordination problem solvable through well-timed nonverbal behaviors, while credibility requires epistemic transparency, pairing behavioral cues with explainability mechanisms (provenance, uncertainty, rationale) that make agent reasoning inspectable. Because our analysis examines naturally occurring rather than experimentally manipulated cues, these findings are associational. Future work should experimentally isolate cue timing and intensity while testing whether transparency overlays can close credibility gaps without compromising cooperativity. Understanding how to design agents that are both engaging collaborators and credible advisors remains essential as embodied AI systems move into complex, dynamic collaborative environments.

References

- [1] Abdullah Alzahrani, Jauwairia Nasir, Ahmad J Tayeb, Elisabeth André, and Muneeb I Ahmad. 2025. What Do the Face and Voice Reveal? Investigating Trust Dynamics During Human-Robot Interaction. In *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction*. 400–409.
- [2] Cigdem Beyan, Alessandro Vinciarelli, and Alessio Del Bue. 2023. Co-located human-human interaction analysis using nonverbal cues: A survey. *Comput. Surveys* 56, 5 (2023), 1–41.
- [3] Judee K Burgoon and Aaron E Bacue. 2003. Nonverbal communication skills. *Handbook of communication and social interaction skills* (2003), 179–219.
- [4] Michael Burgoon. 2002. The persuasion handbook.
- [5] Daniel Buschek, Malin Eiband, and Heinrich Hussmann. 2022. How to support users in understanding intelligent systems? an analysis and conceptual framework of user questions considering user mindsets, involvement, and knowledge outcomes. *ACM Transactions on Interactive Intelligent Systems* 12, 4 (2022), 1–27.
- [6] Fabio Catania, Micol Spitale, and Franca Garzotto. 2023. Conversational agents in therapeutic interventions for neurodevelopmental disorders: a survey. *Comput. Surveys* 55, 10 (2023), 1–34.
- [7] Joyce Chai. 2023. Pragmatic Communication with Embodied Agents. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*. 1–1.
- [8] Andrew J Cowell and Kay M Stanney. 2005. Manipulation of non-verbal interaction style and demographic embodiment to increase anthropomorphic computer character credibility. *International journal of human-computer studies* 62, 2 (2005), 281–306.
- [9] Lee Alan Dugatkin, Michael Mesterton-Gibbons, and Alasdair I Houston. 1992. Beyond the prisoner's dilemma: toward models to discriminate among mechanisms of cooperation in nature. *Trends in ecology & evolution* 7, 6 (1992), 202–205.
- [10] Catalina Gomez, Sue Min Cho, Shichang Ke, Chien-Ming Huang, and Mathias Unberath. 2025. Human-AI collaboration is not very collaborative yet: a taxonomy of interaction patterns in AI-assisted decision making from a systematic review. *Frontiers in Computer Science* Volume 6 - 2024 (2025). doi:10.3389/fcomp.2024.1521066
- [11] Emily Kuang, Ehsan Jahangirzadeh Soure, Mingming Fan, Jian Zhao, and Kristen Shinohara. 2023. Collaboration with conversational AI assistants for UX evaluation: Questions and how to ask them (voice vs. text). In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [12] Yinheng Li, Shaofei Wang, Han Ding, and Hang Chen. 2023. Large language models in finance: A survey. In *Proceedings of the fourth ACM international conference on AI in finance*. 374–382.
- [13] Kate Loveys, Gabrielle Sebaratnam, Mark Sagar, and Elizabeth Broadbent. 2020. The effect of design features on relationship quality with embodied conversational agents: a systematic review. *International Journal of Social Robotics* 12, 6 (2020), 1293–1312.
- [14] James C McCroskey and Thomas J Young. 1981. Ethos and credibility: The construct and its measurement after three decades. *Communication Studies* 32, 1 (1981), 24–34.
- [15] Catharine Oertel, Ginevra Castellano, Mohamed Chetouani, Jauwairia Nasir, Mohammad Obaid, Catherine Pelachaud, and Christopher Peters. 2020. Engagement in human-agent interaction: An overview. *Frontiers in Robotics and AI* 7 (2020), 92.
- [16] OpenAI. 2024. Hello GPT-4o. <https://openai.com/index/hello-gpt-4o/>
- [17] Susanne Schmidt, Gerd Bruder, and Frank Steinicke. 2018. Effects of embodiment on generic and content-specific intelligent virtual agents as exhibition guides. In *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE 2018)*, Limassol, Cyprus, November 7–9, 2018.
- [18] Joanna Schug, David Matsumoto, Yutaka Horita, Toshio Yamagishi, and Kemberlee Bonnet. 2010. Emotional expressivity as a signal of cooperation. *Evolution and Human Behavior* 31, 2 (2010), 87–94.
- [19] Soul Machines. 2024. Soul Machines: Pioneering AGI Research and Humanizing AI Interaction. <https://www.soulmachines.com/> Accessed: 2024-09-02.
- [20] Michelle ME Van Pinxteren, Mark Pluymaekers, and Jos GAM Lemmink. 2020. Human-like communication in conversational agents: a literature review and research agenda. *Journal of Service Management* 31, 2 (2020), 203–225.
- [21] Isaac Wang and Jaime Ruiz. 2021. Examining the use of nonverbal communication in virtual agents. *International Journal of Human-Computer Interaction* 37, 17 (2021), 1648–1673.
- [22] Lawrence R Wheeless and Janis Grotz. 1977. The measurement of trust and its relationship to self-disclosure. *Human Communication Research* 3, 3 (1977), 250–257.
- [23] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2024. Tree of thoughts: Deliberate problem solving with large language models. *Advances in Neural Information Processing Systems* 36 (2024).